

Concept 4 | Higher-Degree Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Factor theorem

$$P(a) = 0 \Rightarrow x - a \text{ is a factor}$$

Degree

$$\deg(PQ) = \deg P + \deg Q$$

Conjugate roots

$$a + bi \Rightarrow a - bi$$

Classified Practice

Higher-Degree Equations

Q001 Written

$$x^3 - 8 = 0$$

Q002 Written

$$x^3 = -1$$

Classified Practice

Higher-Degree Equations

Q003 Written

$$x^3 = 27$$

Q004 Written

$$x^3 = -125$$

Classified Practice

Higher-Degree Equations

Q005 Written

$$8x^3 = 1$$

Q006 MCQ

$x^3 - 8 = 0$ **Select the correct answer:**

Choose the correct option.

- A $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{4}i$
 - B $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$
 - C $x = 2, -1 - \sqrt{3}, -1 + \sqrt{3}i$
 - D $x = 2, -1 - \sqrt{3} - i, -1 + \sqrt{3}i$
-
-
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Classified Practice

Higher-Degree Equations

Q007 MCQ

 $x^3 = -1$ Select the correct answer:

Choose the correct option.

A $x = -1, (1 - \sqrt{3})/2, (1 + \sqrt{3}i)/2$

B $x = -1, (1 - \sqrt{3} - i)/2, (1 + \sqrt{3}i)/2$

C $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/3$

D $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$

Q008 MCQ $x^3 = 27$ **Select the correct answer:****Choose the correct option.**

- A $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/3$
- B $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$
- C $x = 3, (-3 - 3\sqrt{3})/2, (-3 + 3\sqrt{3}i)/2$
- D $x = 3, (-3 - 3\sqrt{3} - i)/2, (-3 + 3\sqrt{3}i)/2$

Classified Practice

Higher-Degree Equations

Q009 MCQ

 $x^3 = -125$ Select the correct answer:

Choose the correct option.

A $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$

B $x = -5, (5 - 5\sqrt{3})/2, (5 + 5\sqrt{3}i)/2$

C $x = -5, (5 - 5\sqrt{3} - i)/2, (5 + 5\sqrt{3}i)/2$

D $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/3$

Q010 MCQ

$8x^3 = 1$ Select the correct answer:

Choose the correct option.

A $x = 1/2, (-1 - \sqrt{3} - i)/4, (-1 + \sqrt{3}i)/4$

B $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/5$

C $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$

D $x = 1/2, (-1 - \sqrt{3})/4, (-1 + \sqrt{3}i)/4$

Classified Practice

Higher-Degree Equations

Q011 Written

$$x^3 - 2x^2 - 5x + 6 = 0$$

Q012 Written

$$3x^3 + x^2 - 8x + 4 = 0$$

Classified Practice

Higher-Degree Equations

Q013 Written

$$2x^3 - 3x^2 - x - 2 = 0$$

Q014 Written

$$x^3 - x^2 - 16x - 20 = 0$$

Classified Practice

Higher-Degree Equations

Q015 Written

$$x^3 + 6x^2 + 11x + 6 = 0$$

Q016 Written

$$x^3 + 3x^2 - 4x - 12 = 0$$

Classified Practice

Higher-Degree Equations

Q017 Written

$$3x^3 - 2x^2 - 3x + 2 = 0$$

Q018 Written

$$x^3 - x^2 - 4 = 0$$

Classified Practice

Higher-Degree Equations

Q019 Written

$$x^3 + 4x^2 + 2x - 4 = 0$$

Q020 Written

$$x^3 - 4x^2 + 9x - 10 = 0$$

Classified Practice

Higher-Degree Equations

Q021 Written

$$x^3 + x^2 - 8x - 12 = 0$$

Q022 Written

$$2x^3 + 7x^2 + 8x + 3 = 0$$

Classified Practice

Higher-Degree Equations

Q023 Written

$$x^3 + 6x^2 + 12x + 8 = 0$$

Q024 MCQ $x^3 - 2x^2 - 5x + 6 = 0$ **Select the correct answer:****Choose the correct option.****A** $x = -2, 1, 3 - 1$ **B** $x = -2, 1, 3$ (sign error)**C** $x = -2, 1, 3$ **D** $x = -2, 1, 4$

Classified Practice

Higher-Degree Equations

Q025 MCQ

 $3x^3 + x^2 - 8x + 4 = 0$ Select the correct answer:

Choose the correct option.

A $x = -2, 2/3, 2$ B $x = -2, 2/3, 1 - 1$ C $x = -2, 2/3, 1$ (sign error)D $x = -2, 2/3, 1$

Q026 MCQ

$2x^3 - 3x^2 - x - 2 = 0$ Select the correct answer:

Choose the correct option.

- A $x = 2, (-1 - \sqrt{7})/4, (-1 + \sqrt{7}i)/4$
 - B $x = 2, (-1 - \sqrt{7} - i)/4, (-1 + \sqrt{7}i)/4$
 - C $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/5$
 - D $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$
-
-
-

Classified Practice

Higher-Degree Equations

Q027 MCQ

 $x^3 - x^2 - 16x - 20 = 0$ Select the correct answer:

Choose the correct option.

A $x = -2$ (repeated), 5 (sign error)B $x = -2$ (repeated), 5C $x = -2$ (repeated), 6D $x = -2$ (repeated), 5 – 1

Q028 MCQ

$x^3 + 6x^2 + 11x + 6 = 0$ Select the correct answer:

Choose the correct option.

- A $x = -3, -2, -1 - 1$
- B $x = -3, -2, -1$ (sign error)
- C $x = -3, -2, -1$
- D $x = -3, -2, 0$

Classified Practice

Higher-Degree Equations

Q029 MCQ

 $x^3 + 3x^2 - 4x - 12 = 0$ Select the correct answer:

Choose the correct option.

A $x = -3, -2, 3$ B $x = -3, -2, 2 - 1$ C $x = -3, -2, 2$ (sign error)D $x = -3, -2, 2$

Q030 MCQ $3x^3 - 2x^2 - 3x + 2 = 0$ **Select the correct answer:****Choose the correct option.****A** $x = -1, 2/3, 2$ **B** $x = -1, 2/3, 1 - 1$ **C** $x = -1, 2/3, 1$ (sign error)**D** $x = -1, 2/3, 1$

Classified Practice

Higher-Degree Equations

Q031 MCQ

 $x^3 - x^2 - 4 = 0$ Select the correct answer:

Choose the correct option.

A $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$

B $x = 2, (-1 - \sqrt{7})/2, (-1 + \sqrt{7}i)/2$

C $x = 2, (-1 - \sqrt{7} - i)/2, (-1 + \sqrt{7}i)/2$

D $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/3$

Classified Practice

Higher-Degree Equations

Q032 MCQ

 $x^3 + 4x^2 + 2x - 4 = 0$ Select the correct answer:

Choose the correct option.

A $x = -2, -1 - \sqrt{3}, -1 + \sqrt{4}$

B $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3} - 1$

C $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$ (sign error)

D $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$

Classified Practice

Higher-Degree Equations

Q033 MCQ

 $x^3 - 4x^2 + 9x - 10 = 0$ Select the correct answer:

Choose the correct option.

A $x = 2, 1 - 2, 1 + 2i$

B $x = 2, 1 - 2 - i, 1 + 2i$

C $x = 2, 1 - 2i, 1 + 3i$

D $x = 2, 1 - 2i, 1 + 2i$

Q034 MCQ

$x^3 + x^2 - 8x - 12 = 0$ Select the correct answer:

Choose the correct option.

A $x = -2$ (repeated), 4

B $x = -2$ (repeated), 3 – 1

C $x = -2$ (repeated), 3 (sign error)

D $x = -2$ (repeated), 3

Classified Practice

Higher-Degree Equations

Q035 MCQ

 $2x^3 + 7x^2 + 8x + 3 = 0$ Select the correct answer:

Choose the correct option.

A $x = -1$ (repeated), $-3/2$ (sign error)B $x = -1$ (repeated), $-3/2$ C $x = -1$ (repeated), $-3/2 + 1$ D $x = -1$ (repeated), $-3/2 - 1$

Q036 MCQ

$x^3 + 6x^2 + 12x + 8 = 0$ **Select the correct answer:**

Choose the correct option.

- A $x = -1$ (triple)
 - B $x = -2$ (triple) – 1
 - C $x = -2$ (triple) (sign error)
 - D $x = -2$ (triple)
-
-
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Classified Practice

Higher-Degree Equations

Q037 Written

$$x^4 - 7x^2 - 18 = 0$$

Q038 Written

$$x^4 - 5x^2 + 4 = 0$$

Classified Practice

Higher-Degree Equations

Q039 Written

$$x^4 - 5x^2 - 36 = 0$$

Q040 Written

$$x^4 - 6x^2 - 27 = 0$$

Classified Practice

Higher-Degree Equations

Q041 Written

$$x^4 - x^2 - 2 = 0$$

Q042 MCQ $x^4 - 7x^2 - 18 = 0$ **Select the correct answer:****Choose the correct option.**

- A $x = -3, 3, -\sqrt{2}, \sqrt{2}i$
B $x = -3, 3, -\sqrt{2} - i, \sqrt{2}i$
C $x = -3, 3, -\sqrt{2}i, \sqrt{3}i$
D $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$
-
-
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Classified Practice

Higher-Degree Equations

Q043 MCQ

 $x^4 - 5x^2 + 4 = 0$ Select the correct answer:

Choose the correct option.

A $x = -2, -1, 1, 3$ B $x = -2, -1, 1, 2 - 1$ C $x = -2, -1, 1, 2$ (sign error)D $x = -2, -1, 1, 2$

Q044 MCQ $x^4 - 5x^2 - 36 = 0$ **Select the correct answer:****Choose the correct option.****A** $x = -3, 3, -2 - i, 2i$ **B** $x = -3, 3, -2i, 3i$ **C** $x = -3, 3, -2i, 2i$ **D** $x = -3, 3, -2, 2i$

Classified Practice

Higher-Degree Equations

Q045 MCQ

 $x^4 - 6x^2 - 27 = 0$ Select the correct answer:

Choose the correct option.

A $x = -3, 3, -\sqrt{3} - i, \sqrt{3}i$

B $x = -3, 3, -\sqrt{3}i, \sqrt{4}i$

C $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$

D $x = -3, 3, -\sqrt{3}, \sqrt{3}i$

Q046 MCQ $x^4 - x^2 - 2 = 0$ **Select the correct answer:****Choose the correct option.**

A $x = -\sqrt{2}, \sqrt{3}, -i, i$

B $x = -\sqrt{2}, \sqrt{2}, -i, i$

C $x = -\sqrt{2}, \sqrt{2}, -, i$

D $x = -\sqrt{2}, \sqrt{2}, +i, i$

Classified Practice

Higher-Degree Equations

Q047 Written

$$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$$

Q048 Written

$$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$$

Classified Practice

Higher-Degree Equations

Q049 Written

$$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$$

Q050 Written

$$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$$

Classified Practice

Higher-Degree Equations

Q051 MCQ

 $x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$ Select the correct answer:

Choose the correct option.

A $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$ (sign error)B $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$ C $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{6}$ D $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5} - 1$

Classified Practice

Higher-Degree Equations

Q052 MCQ

 $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ Select the correct answer:

Choose the correct option.

A $x = 1, 2, 3, 4$ B $x = 1, 2, 3, 5$ C $x = 1, 2, 3, 4 - 1$ D $x = 1, 2, 3, 4$ (sign error)

Q053 MCQ $x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$ **Select the correct answer:****Choose the correct option.****A** $x = 1, 2, 3(\text{repeated}) - 1$ **B** $x = 1, 2, 3(\text{repeated})$ (sign error)**C** $x = 1, 2, 3(\text{repeated})$ **D** $x = 1, 2, 4(\text{repeated})$

Classified Practice

Higher-Degree Equations

Q054 MCQ

 $x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$ Select the correct answer:

Choose the correct option.

A $x = -1, 3, 1 - i, 1 + i$

B $x = -1, 3, 1 - i, 2 + i$

C $x = -1, 3, 1 - i, 1 + i$

D $x = -1, 3, 1 - i, 1 + i$

Q055 Written

$x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root.

Classified Practice

Higher-Degree Equations

Q056 Written

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a,b and the other root.

Q057 Written

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a,b and the other root.

Classified Practice

Higher-Degree Equations

Q058 MCQ

$x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root Select the correct answer:

Choose the correct option.

A $a = 1, b = -6$; other roots: $-2 - 1$

B $a = 1, b = -6$; other roots: -2 (sign error)

C $a = 1, b = -6$; other roots: -2

D $a = 1, b = -6$; other roots: -1

Classified Practice

Higher-Degree Equations

Q059 MCQ

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a,b and the other root Select the correct answer:

Choose the correct option.

A $a = 2, b = 2$; other roots: 1 (sign error)

B $a = 2, b = 2$; other roots: 1

C $a = 2, b = 2$; other roots: 2

D $a = 2, b = 2$; other roots: 1 - 1

Classified Practice

Higher-Degree Equations

Q060 MCQ

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a,b and the other root Select the correct answer:

Choose the correct option.

A $a = -7, b = 14$; other roots: $4 - 1$

B $a = -7, b = 14$; other roots: 4 (sign error)

C $a = -7, b = 14$; other roots: 4

D $a = -7, b = 14$; other roots: 5

Classified Practice

Higher-Degree Equations

Q061 Written

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots.

Q062 Written

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots.

Classified Practice

Higher-Degree Equations

Q063 Written

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots.

Q064 Written

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots.

Classified Practice

Higher-Degree Equations

Q065 MCQ

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots Select the correct answer:

Choose the correct option.

A $a = -3, b = 1$; other roots: $-1, 2 - i$

B $a = -3, b = 1$; other roots: $-1, 2 -$

C $a = -3, b = 1$; other roots: $-1, 2 + i$

D $a = -3, b = 1$; other roots: $-1, 3 - i$

Q066 MCQ

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots Select the correct answer:

Choose the correct option.

- A $a = -4, b = 6$; other roots: $1 - i, 3$
- B $a = -4, b = 6$; other roots: $1 - i, 2$
- C $a = -4, b = 6$; other roots: $1-, 2$
- D $a = -4, b = 6$; other roots: $1 + i, 2$

Classified Practice

Higher-Degree Equations

Q067 MCQ

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots Select the correct answer:

Choose the correct option.

- A $a = -4, b = 14$; other roots: $1 + 3, 2$
B $a = -4, b = 14$; other roots: $1 + 3 - i, 2$
C $a = -4, b = 14$; other roots: $1 + 3i, 3$
D $a = -4, b = 14$; other roots: $1 + 3i, 2$

Q068 MCQ

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots Select the correct answer:

Choose the correct option.

A $a = 0, b = 1$; other roots: $1 - 2i, -1$

B $a = 0, b = 1$; other roots: $1 - 2i, -2$

C $a = 0, b = 1$; other roots: $1 - 2, -2$

D $a = 0, b = 1$; other roots: $1 - 2 - i, -2$

Classified Practice

Higher-Degree Equations

Q069 Written

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots.

Q070 MCQ

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots Select the correct answer:

Choose the correct option.

A $k = 9$; other roots: 3, 4 (sign error)

B $k = 9$; other roots: 3, 4

C $k = 9$; other roots: 3, 5

D $k = 9$; other roots: 3, 4 - 1

Classified Practice

Higher-Degree Equations

Q071 Written

For an imaginary cube root of unity w , evaluate w^9 .

Q072 Written**For an imaginary cube root of unity w , evaluate w^{2010} .**

Classified Practice

Higher-Degree Equations

Q073 **Written**

For an imaginary cube root of unity w , evaluate w^3 .

Q074 MCQ

For an imaginary cube root of unity w , evaluate w^9 Select the correct answer:

Choose the correct option.

A 1

B 2

C $1 - 1$

D 1 (sign error)

Classified Practice

Higher-Degree Equations

Q075 MCQ

For an imaginary cube root of unity w , evaluate w^{2010} Select the correct answer:

Choose the correct option.

A 1 (sign error)

B 1

C 2

D $1 - 1$

Q076 MCQ**For an imaginary cube root of unity w , evaluate w^3 Select the correct answer:****Choose the correct option.****A** 2**B** $1 - 1$ **C** 1 (sign error)**D** 1

Classified Practice

Higher-Degree Equations

Q077 Written

For an imaginary cube root of unity w , evaluate $w^4 + w^5$.

Q078 Written**For an imaginary cube root of unity w , evaluate $w^2 + w$.**

Classified Practice

Higher-Degree Equations

Q079 Written

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$.

Q080 MCQ**For an imaginary cube root of unity w , evaluate $w^4 + w^5$ Select the correct answer:****Choose the correct option.****A** -1 **B** 0 **C** $-1 - 1$ **D** -1 (sign error)

Classified Practice

Higher-Degree Equations

Q081 MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

Choose the correct option.

A -1 (sign error)

B -1

C 0

D $-1 - 1$

Q082 MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$ Select the correct answer:

Choose the correct option.

A 1

B $0 - 1$

C 0 (sign error)

D 0

Classified Practice

Higher-Degree Equations

Q083 Written

For an imaginary cube root of unity w , evaluate $w + 1/w$.

Q084 Written

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.

Classified Practice

Higher-Degree Equations

Q085 Written

For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$.

Q086 MCQ**For an imaginary cube root of unity w , evaluate $w + 1/w$ Select the correct answer:****Choose the correct option.****A** 0**B** $-1 - 1$ **C** -1 (sign error)**D** -1

Classified Practice

Higher-Degree Equations

Q087 MCQ

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$ Select the correct answer:

Choose the correct option.

A $1 - 1$

B 1 (sign error)

C 1

D 2

Q088 MCQ**For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$ Select the correct answer:****Choose the correct option.****A** 1**B** $0 - 1$ **C** 0 (sign error)**D** 0

Classified Practice

Higher-Degree Equations

Q089 Written

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root.

Q090 Written

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root.

Classified Practice

Higher-Degree Equations

Q091 **Written**

$x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root.

Q092 Written

$x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root.

Classified Practice

Higher-Degree Equations

Q093 Written

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root.

Q094 MCQ

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Choose the correct option.

A $a = -4$ or $9/4 + 1$

B $a = -4$ or $9/4 - 1$

C $a = -4$ or $9/4$ (sign error)

D $a = -4$ or $9/4$

Classified Practice

Higher-Degree Equations

Q095 MCQ

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Choose the correct option.

A $a = -3$ or 2

B $a = -3$ or $1 - 1$

C $a = -3$ or 1 (sign error)

D $a = -3$ or 1

Q096 MCQ

$x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Choose the correct option.

A $a = 5$ or 9 (sign error)

B $a = 5$ or 9

C $a = 5$ or 10

D $a = 5$ or $9 - 1$

Classified Practice

Higher-Degree Equations

Q097 MCQ

$x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Choose the correct option.

A $a = -6, -2$, or 10

B $a = -6, -2$, or 11

C $a = -6, -2$, or $10 - 1$

D $a = -6, -2$, or 10 (sign error)

Q098 MCQ

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Choose the correct option.

A $a = -7, -3$, or 2

B $a = -7, -3$, or 3

C $a = -7, -3$, or $2 - 1$

D $a = -7, -3$, or 2 (sign error)

Classified Practice

Higher-Degree Equations

Q099 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots.

Classified Practice

Higher-Degree Equations

Q100 Challenge

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion.

Classified Practice

Higher-Degree Equations

Q101 Challenge

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root.

Classified Practice

Higher-Degree Equations

Q102 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$.

Classified Practice

Higher-Degree Equations

Q103 Challenge**Solve $x^4 - x^3 + x^2 + x - 2 = 0$ for the stable pressure levels.**

Classified Practice

Higher-Degree Equations

Q104 **Challenge****Classify the stable pressure solutions as real or complex.**

Classified Practice

Higher-Degree Equations

Q105 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots Select the correct answer:

Choose the correct option.

A $a = -1, b = 6$; other roots: $1 - \sqrt{2} - i, -2$

B $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -1$

C $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$

D $a = -1, b = 6$; other roots: $1 - \sqrt{2}, -2$

Classified Practice

Higher-Degree Equations

Q106 Challenge

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion. Select the correct answer:

Choose the correct option.

A 5

B $4 - 1$

C 4 (sign error)

D 4

Classified Practice

Higher-Degree Equations

Q107 Challenge

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Choose the correct option.

A $a = 2/3, 2$, or 7

B $a = 2/3, 2$, or $6 - 1$

C $a = 2/3, 2$, or 6 (sign error)

D $a = 2/3, 2$, or 6

Classified Practice

Higher-Degree Equations

Q108 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$ Select the correct answer:

Choose the correct option.

A $x^4 - x^3 + x^2 + x - 2 = 0$

B $x^4 - x^3 + x^2 + x - 2 = 1$

C $x^4 - x^3 + x^2 + x - 2 = 0 - 1$

D $x^4 - x^3 + x^2 + x - 2 = 0$ (sign error)

Classified Practice

Higher-Degree Equations

Q109 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ **for the stable pressure levels** **Select the correct answer:**

Choose the correct option.

A $x = -1, 1, (1 - \sqrt{7})/2, (1 + \sqrt{7}i)/2$

B $x = -1, 1, (1 - \sqrt{7} - i)/2, (1 + \sqrt{7}i)/2$

C $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/3$

D $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$

Classified Practice

Higher-Degree Equations

Q110 Challenge

Classify the stable pressure solutions as real or complex Select the correct answer:

Choose the correct option.

- A Both types occur : $\text{real } x = -1, 1$ and $\text{non-real conjugates } x = (1 \pm \sqrt{7}i)/2$. If pressure is restricted to $x \geq 0$, the real physical level is $x = 2$.
- B Both types occur : $\text{real } x = -1, 1$ and $\text{non-real conjugates } x = (1 \pm \sqrt{7}i)/2$. If pressure is restricted to $x \geq 0$, the real physical level is $x = 1$.
- C Both types occur : $\text{real } x = -1, 1$ and $\text{non-real conjugates } x = (1 \pm \sqrt{7})/2$. If pressure is restricted to $x \geq 0$, the real physical level is $x = 1$.
- D Both types occur : $\text{real } x = -1, 1$ and $\text{non-real conjugates } x = (1 \pm \sqrt{7} - i)/2$. If pressure is restricted to $x \geq 0$, the real physical level is $x = 1$.

Classified Practice

Higher-Degree Equations

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $x^3 - 8 = 0$. Select the correct answer:

Choose the correct option.

A $x = 2, -1 - \sqrt{3}, -1 + \sqrt{3}i$

B $x = 2, -1 - \sqrt{3} - i, -1 + \sqrt{3}i$

C $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{4}i$

D $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$

Quiz MCQ 2 Concept Quiz · MCQ

Are a cube root of $1 + 2i$. Which root must also occur? Select the correct answer:

Choose the correct option.

A $1 - 2$

B $1 - 2 - i$

C $1 - 1i$

D $1 - 2i$

Classified Practice

Higher-Degree Equations

Quiz MCQ 3 [Concept Quiz](#) · MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

Choose the correct option.

A -1 (sign error)

B -1

C 0

D $-1 - 1$

Quiz MCQ 4 Concept Quiz · MCQ

For the remaining quadratic factor, a repeated root requires Select the correct answer:

Choose the correct option.

A $D = 0$ (sign error)

B $D = 0$

C $D = 1$

D $D = 0 - 1$

Classified Practice

Higher-Degree Equations

Quiz WRITTEN 5 Concept Quiz · Written

Solve $x^4 - 5x^2 + 4 = 0$.

Quiz WRITTEN 6 Concept Quiz · Written

For $x^3 + ax^2 + bx + 10 = 0$ with root $1 + 2i$, find a, b and the other roots.

Classified Practice

Higher-Degree Equations

Quiz WRITTEN 7 Concept Quiz · Written

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.

Quiz WRITTEN 8 Concept Quiz · Written

Find all real a for $x^3 - 2x^2 + (a - 3)x + a = 0$ to have a repeated root.

Answer key

Use the question number shown on each page.

Q001 $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$	Q027 B	Q053 C	Q079 0
Q002 $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$	Q028 C	Q054 C	Q080 A
Q003 $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$	Q029 D	Q055 $a = 1, b = -6$; other roots: -2	Q081 B
Q004 $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$	Q030 D	Q056 $a = 2, b = 2$; other roots: 1	Q082 D
Q005 $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$	Q031 A	Q057 $a = -7, b = 14$; other roots: 4	Q083 -1
Q006 B	Q032 D	Q058 C	Q084 1
Q007 D	Q033 D	Q059 B	Q085 0
Q008 B	Q034 D	Q060 C	Q086 D
Q009 A	Q035 B	Q061 $a = -3, b = 1$; other roots: $-1, 2 - i$	Q087 C
Q010 C	Q036 D	Q062 $a = -4, b = 6$; other roots: $1 - i, 2$	Q088 D
Q011 $x = -2, 1, 3$	Q037 $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$	Q063 $a = -4, b = 14$; other roots: $1 + 3i, 2$	Q089 $a = -4$ or $9/4$
Q012 $x = -2, 2/3, 1$	Q038 $x = -2, -1, 1, 2$	Q064 $a = 0, b = 1$; other roots: $1 - 2i, -2$	Q090 $a = -3$ or 1
Q013 $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$	Q039 $x = -3, 3, -2i, 2i$	Q065 A	Q091 $a = 5$ or 9
Q014 $x = -2(\text{repeated}), 5$	Q040 $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$	Q066 B	Q092 $a = -6, -2$, or 10
Q015 $x = -3, -2, -1$	Q041 $x = -\sqrt{2}, \sqrt{2}, -i, i$	Q067 D	Q093 $a = -7, -3$, or 2
Q016 $x = -3, -2, 2$	Q042 D	Q068 B	Q094 D
Q017 $x = -1, 2/3, 1$	Q043 D	Q069 $k = 9$; other roots: $3, 4$	Q095 D
Q018 $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$	Q044 C	Q070 B	Q096 B
Q019 $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$	Q045 C	Q071 1	Q097 A
Q020 $x = 2, 1 - 2i, 1 + 2i$	Q046 B	Q072 1	Q098 A
Q021 $x = -2(\text{repeated}), 3$	Q047 $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$	Q073 1	Q099 $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$
Q022 $x = -1(\text{repeated}), -3/2$	Q048 $x = 1, 2, 3, 4$	Q074 A	Q100 4
Q023 $x = -2(\text{triple})$	Q049 $x = 1, 2, 3(\text{repeated})$	Q075 B	Q101 $a = 2/3, 2$, or 6
Q024 C	Q050 $x = -1, 3, 1 - i, 1 + i$	Q076 D	Q102 $x^4 - x^3 + x^2 + x - 2 = 0$
Q025 D	Q051 B	Q077 -1	Q103 $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$
Q026 D	Q052 A	Q078 -1	

Both types occur: real $x = -1, 1$		Quiz MCQ 1	Quiz WRITTEN 5
Q104 non-real conjugates $x = (1 \pm \sqrt{7}i)/2$	Q107 D	D	$x = -2, -1, 1, 2$
if $x \geq 0 : x = 1$	Q108 A	Quiz MCQ 2	Quiz WRITTEN 6
Q105 C	Q109 D	B	$a = 0, b = 1$; other roots: $1 - 2i, -2$
Q106 D	Q110 B	Quiz MCQ 3	Quiz WRITTEN 7
		B	1
		Quiz MCQ 4	Quiz WRITTEN 8
		B	$a = -4$ or $9/4$